Unit 2 Pre-Test

Scientific Notation

Standards assessed: NC.8.EE.3, NC.8.EE.4

Name: _____ **Date:** _____ **Period:** _____

SCORE **3.0** **Basic Understanding**

NC.8.EE.3

• I can write numbers in scientific notation and in standard form, like 45,000 = 4.5 × 10^4.

Match each vocabulary term to its definition, fill in the parts of a number in scientific notation, then write each number as directed.

D scientific notation ✓	**A.** A number written the ordinary way. *Example: 2,700 or 0.05*
B coefficient ✓	**B.** The decimal factor in a × 10^n — at least 1 and less than 10. *Example: the 8.3 in 8.3 × 10^6*
A standard form ✓	**C.** Shows a small number between 0 and 1 (move the decimal left). *Example: the −2 in 5 × 10^{-2} = 0.05*
E base ✓	**D.** A number written as a × 10^n, where 1 ≤ a < 10. *Example: 8.3 × 10^6*
F positive exponent ✓	**E.** In a power of ten, the number that is multiplied repeatedly. *Example: the 10 in 10^4 = 10 · 10 · 10 · 10*
C negative exponent ✓	**F.** Shows a large number, 10 or greater (move the decimal right). *Example: the 6 in 8.3 × 10^6 = 8,300,000*

Parts of a number in scientific notation — fill in the blanks

$$3.4 \times 10^{5}$$

coefficient	base	exponent
the 3.4	the 10	the 5

A positive exponent makes a **large** *number; a negative exponent makes a* **small** *one.*

1. *Write in scientific notation*

45,000

Answer: _____

answer: 4.5×10^{4}

2. *Write in standard form*

3×10^{4}

Answer: _____

answer: 30,000

3. *Write in scientific notation*

0.0062

Answer: _____

answer: 6.2×10^{-3}

4. *Write in standard form*

7.1×10^{-2}

Answer: _____

answer: 0.071

5. *Write in scientific notation*

580,000

Answer: _____

answer: 5.8×10^{5}

6. *Fix the notation*

Is 12.5×10^{6} correct? Write the correct form.

Answer: _____

answer: 1.25×10^{7}

SCORE 4.0 Target Content

NC.8.EE.4

- *I can multiply and divide in scientific notation and say how many times bigger one number is.*

Multiply, divide, or compare. Write each answer in correct scientific notation.

7. *Multiply*

$(3 \times 10^4)(2 \times 10^3)$

Answer:

answer: 6×10^7

8. *Divide*

$(8 \times 10^9) \div (4 \times 10^5)$

Answer:

answer: 2×10^4

9. *Compare*

A blue whale has a mass of about 1.5×10^5 kg. A dog has a mass of about 3×10^1 kg. How many times greater is the whale's mass?

Answer:

answer: 5×10^3 (5,000 times)

10. *Compare*

How many times larger is 6×10^8 than 2×10^3?

Answer:

answer: 3×10^5

11. *Divide*

$(9 \times 10^{-2}) \div (3 \times 10^4)$

Answer:

answer: 3×10^{-6}

12. *Multiply*

$(2.5 \times 10^5)(2 \times 10^{-8})$

Answer:

answer: 5×10^{-3}

SCORE **5.0** **Advanced Content**

NC.8.EE.4

• I can multiply and divide with negative exponents and solve word problems in scientific notation.

Solve. Show your work. Write each answer in correct scientific notation (the first number must be at least 1 and less than 10).

13. *Multiply, then fix the form*

$(4 \times 10^{-3})(6 \times 10^{-5})$

Answer:

answer: $24 \times 10^{-8} \rightarrow 2.4 \times 10^{-7}$

14. *Divide, then fix the form*

$(1.2 \times 10^{-4}) \div (4 \times 10^{-9})$

Answer:

answer: $0.3 \times 10^{5} \rightarrow 3 \times 10^{4}$

15. *Multi-step, then fix the form*

$$\frac{(7.5 \times 10^{-6})(4 \times 10^{3})}{2.5 \times 10^{-4}}$$

Answer:

answer: $12 \times 10^{1} \rightarrow 1.2 \times 10^{2}$

16. *Apply*

A red blood cell is about 8×10^{-6} meter wide. A grain of sand is about 6.4×10^{-4} meter wide. How many times wider is the grain of sand than the red blood cell? Write your answer in scientific notation.

Answer:

answer: $0.8 \times 10^{2} \rightarrow 8 \times 10^{1}$ (80 times)

17. *Apply*

One bacterium has a mass of about 2×10^{-12} gram. A sample has 6×10^{8} bacteria. What is the total mass of the sample, in grams?

Answer:

answer: $12 \times 10^{-4} \rightarrow 1.2 \times 10^{-3}$ g

18. *Apply*

A hard drive holds 2×10^{12} bytes. One song file uses about 4×10^{6} bytes. How many song files can the hard drive hold?

Answer:

answer: $0.5 \times 10^{6} \rightarrow 5 \times 10^{5}$ songs